

Logistic回归分析-分类

- logistic回归是从统计学中借鉴来的，尽管名字里有回归俩字儿，但它**不是**一个需要**预测连续结果的回归算法**。与之相反，Logistic 回归是**二分类任务**的首选方法。
- 它输出一个 0 到 1 之间的离散二值结果。简单来说，它的结果不是 1 就是 0。



Logistic 回归：概率型回归

用于分析某类事件发生的概率与自变量之间的关系。适用于因变量是分类变量的资料，尤其是二分类的情形。

注：线性回归：因变量是连续型变量



Logistic回归分析解决的问题

医学研究中，有关**生存与死亡**，**发病与未发病**，**阴性与阳性**等结果的产生，可能与病人的年龄、性别、生活习惯、体质、遗传、心理等许多因素有关。如何找出其中**哪些因素**对结果有影响？以及**影响有多大**？

分析一个因变量（分类变量）与多个自变量之间的关系



实例1

- 某研究者调查了30名成年人，记录了同肺癌发病的有关因素情况，数据见表4。其中**是否患病**中，‘0’代表否，‘1’代表是；**性别**中‘1’代表男，‘0’代表女；**吸烟**中‘1’代表吸烟，‘0’代表不吸烟；**地区**中，‘1’代表农村，‘0’代表城市。试分析各因素与**肺癌**间的关系。

表 肺癌与危险因素的调查分析

例号	是否患病	性别	吸烟	年龄	地区
1	1	1	0	30	0
2	1	0	1	46	1
3	0	0	0	35	1
...	...	...	...	...	...
30	0	0	0	26	1

注：是否患病中，‘0’代表否，‘1’代表是。性别中‘1’代表男，‘0’代表女，吸烟中‘1’代表吸烟，‘0’代表不吸烟。地区中，‘1’代表农村，‘0’代表城市。



Variables in the Equation

自变量对A发生概率
是否有显著影响

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	X1	2.520	1.821	1.916	1	.166	12.431
	X2	3.999	1.975	4.101	1	.043	54.568
	X3	.189	.079	5.708	1	.017	1.208
	X4	-1.306	1.583	.681	1	.409	.271
	Constant	-9.781	4.099	5.694	1	.017	.000

a. Variable(s) entered on step 1: X1, X2, X3, X4.

□ 由上最大似然估计分析知因素X2(吸烟), X3(年龄)对肺癌的发生有影响。 所得的回归方程为:

$$\text{Logit}(P) = -9.7544 + 2.5152X_1 + 3.9849X_2 + 0.1884X_3 - 1.3037X_4$$

Classification Table

Observed			Predicted		
			Y		Percentage Correct
			0	1	
Step 1	Y	0	14	1	93.3
		1	2	13	86.7
Overall Percentage					90.0

a. The cut value is .500



Logistic回归分析-分类

logistic回归是一种广义的线性回归分析模型，常用于数据挖掘，疾病自动诊断，经济预测等领域。例如，探讨引发疾病的危险因素，并根据危险因素预测疾病发生的概率等。

对于随机事件A，发生与否用随机变量Y表示，则

$$Y = \begin{cases} 1, & \text{若} A \text{发生} \\ 0, & \text{若} A \text{不发生} \end{cases}$$



工作原理

- Logistic 回归通过使用其固有的 **logistic 函数估计概率**，来衡量因变量（我们想要预测的标签）与一个或多个自变量（特征）之间的关系。
- 然后这些概率必须**二值化**才能真地进行预测。这就是 logistic 函数的任务，也称为 **sigmoid 函数**。Sigmoid 函数是一个 S 形曲线，它可以将任意实数值映射到介于 0 和 1 之间的值，但并不会取到 0/1。然后使用阈值分类器将 0 和 1 之间的值转换为 0 或 1。



Logistic回归分析-分类

事件A发生的概率

$$P(A)=P(Y=1)=E(Y)$$

假设影响A事件发生的因素有p-1个，记为 $X_1, X_1, \dots, X_{p-1}$

Logistic回归分析的目的：

建立事件A发生的概率 $P(Y=1)$ 与 $X_1, X_1, \dots, X_{p-1}$ 之间的关系模型

即：

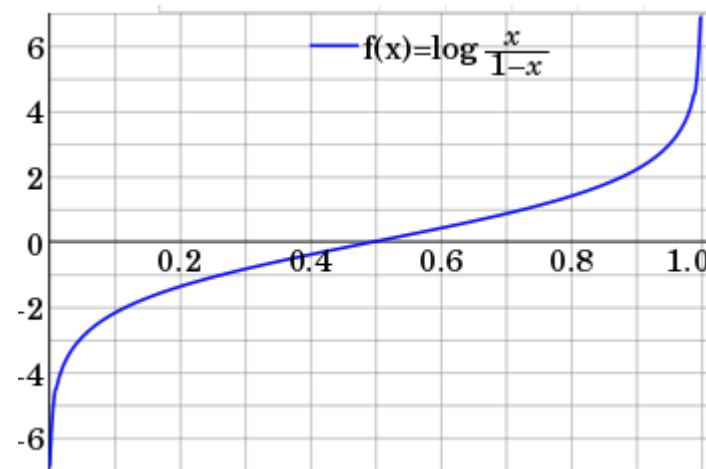
$$E(Y) = \beta_0 + \beta_1 X_1 + \dots + \beta_{p-1} X_{p-1} = \beta_0 + \sum_{j=1}^{p-1} \beta_j X_j$$



Logistic回归分析

Logit函数变换

$$\text{Logit}(E(Y)) = \ln\left(\frac{E(Y)}{1-E(Y)}\right)$$



得到Logistic回归模型:

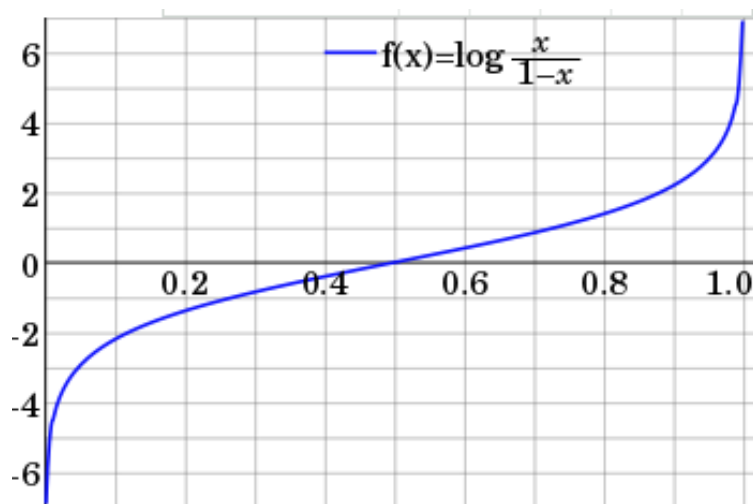
$$\ln\left(\frac{E(Y)}{1-E(Y)}\right) = \beta_0 + \sum_{j=1}^p \beta_j X_j$$

或

$$P(Y=1) = E(Y) = \frac{\exp\left(\beta_0 + \sum_{j=1}^p \beta_j X_j\right)}{1 + \exp\left(\beta_0 + \sum_{j=1}^p \beta_j X_j\right)}$$
$$= \frac{1}{1 + e^{-\left(\beta_0 + \sum_{j=1}^p \beta_j X_j\right)}} = \text{Sigmoid}\left(\beta_0 + \sum_{j=1}^p \beta_j X_j\right)$$

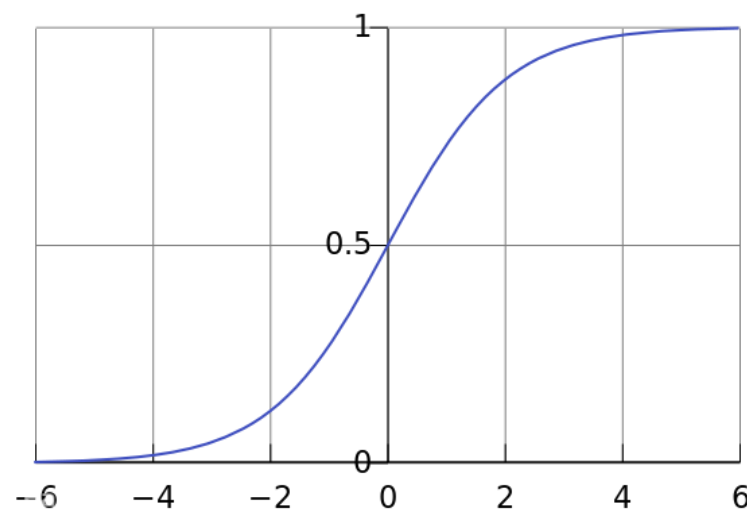


Logistic函数, Siomod函数



Logit函数

$$f(x) = \frac{1}{1 + e^{-x}}$$



Logistic/Sigmoid函数



2、Logistic 回归模型的基本结构

设 $X_1, X_2, \dots, X_p$ 是一组自变量， Y 是因变量（**阳性记为 $y = 1$ ，阴性记为 $y = 0$** ），用 P 表示发生**阳性结果**的概率。则：

$$P = \frac{\exp(\beta_0 + \beta_1 X_1 + L + \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + L + \beta_p X_p)}$$

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + L + \beta_p X_p)}}$$

Logistic 函数

Logistic回归分析

假设自变量为 $X_1, X_1, \dots, X_{p-1}$

对事件A观测了n次，第k次的观测值为

$$X_k = (X_{k,1}, X_{k,2}, \dots, X_{k,p-1}), 1 \leq k \leq n$$

设自变量取 X_k 时，事件A发生的概率为 p_k

则，事件A发生的概率服从二项分布，也就是

$$Y \sim b(n, p_k), k = 1, 2, \dots, n$$



Logistic回归分析

Logistic回归模型

$$\ln\left(\frac{p_k}{1-p_k}\right) = \beta_0 + \sum^k \beta X_k$$

使用最大似然法对Logistic回归模型进行回归参数 β 估计，最终得到结果：

在观测值为 X_k 时，事件A发生的概率

$$p_k = \frac{\exp(\beta_0 + \sum^k \beta X_k)}{1 + \exp(\beta_0 + \sum^k \beta X_k)}$$



Logistic回归分析模型的统计推断

--自变量显著性检验

重要检验：

1. 全局回归关系的显著性；  一组自变量的似然比检验
2. 各自变量对A发生概率是否有显著影响；  Wald检验

1. 一组自变量的似然比检验

将所有 $p - 1$ 个自变量分为两组，一组含 r 个，另一组含 $p - r - 1$ 个. 为便于叙述，不妨设这两组自变量分别为 $X_1, X_2, \dots, X_r$ 和 $X_{r+1}, X_{r+2}, \dots, X_{p-1}$ ，检验 $X_1, X_2, \dots, X_r$ 对事件 A 发生的概率影响的显著性，即检验假设

$$\underline{H_0: \beta_1 = \beta_2 = \dots = \beta_r = 0} \leftrightarrow H_1: \beta_1, \beta_2, \dots, \beta_r \text{ 中至少有一个不为零.}$$



一组自变量的似然比检验

$$\begin{aligned} L(H_1) &= \ln L(\hat{\boldsymbol{\beta}}; y_1, y_2, \dots, y_m) \\ &= \sum_{k=1}^m \ln \binom{n_k}{y_k} - \sum_{k=1}^m n_k \ln [1 + \exp(\tilde{\mathbf{x}}_k^T \hat{\boldsymbol{\beta}})] + \sum_{k=1}^m y_k \tilde{\mathbf{x}}_k^T \hat{\boldsymbol{\beta}}. \end{aligned} \quad (2.91)$$

含所有
自变量

$$\begin{aligned} L(H_0) &= \ln L(\hat{\boldsymbol{\beta}}_{H_0}; y_1, y_2, \dots, y_m) \\ &= \sum_{k=1}^m \ln \binom{n_k}{y_k} - \sum_{k=1}^m n_k \ln [1 + \exp((\tilde{\mathbf{x}}_k^{(0)})^T \hat{\boldsymbol{\beta}}_{H_0})] + \sum_{k=1}^m y_k (\tilde{\mathbf{x}}_k^{(0)})^T \hat{\boldsymbol{\beta}}_{H_0}, \end{aligned} \quad (2.92)$$

$\beta_1 \dots \beta_r$
 $= 0$

$$\tilde{\mathbf{x}}_k^{(0)} = (1, x_{k,r+1}, \dots, x_{k,p-1}), k = 1, 2, \dots, m.$$

构造统计量
 K^2

$$K^2 = 2[L(H_1) - L(H_0)] = 2 \ln \left[\frac{L(\hat{\boldsymbol{\beta}}; y_1, y_2, \dots, y_m)}{L(\hat{\boldsymbol{\beta}}_{H_0}; y_1, y_2, \dots, y_m)} \right] \quad (2.93)$$

$$p_0 = P_{H_0}(K^2 \geq K_0^2) = P(\chi^2(r) \geq K_0^2), \quad (2.94)$$



Wald检验

□ 各自变量对A发生概率是否有显著影响检验

$$H_0: \beta_j = 0 \leftrightarrow H_1: \beta_j \neq 0.$$

$$\frac{\hat{\beta}_j - \beta_j}{s(\hat{\beta}_j)} \underset{A}{\sim} N(0, 1). \quad (2.96)$$

正态分布

若 $\beta_j=0$; 可得 W_j 满足 $\chi^2(1)$ 分布

$$W_j = \left(\frac{\hat{\beta}_j}{s(\hat{\beta}_j)} \right)^2, \quad (2.97)$$

构造统计量

$$P_{0j} = P_{H_0}(W_j \geq W_{0j}) = P(\chi^2(1) \geq W_{0j}), \quad (2.98)$$

例 2.11

- 40位肺癌患者在六个变量上的观测值：生活能力评分 X_1 、年龄 X_2 、确诊到进入观测研究的时间 X_3 、肿瘤的类型 X_4 、化疗方法 X_5 ；患者的生存时间 >200 天： Y

表 2.12 40 位肺癌患者的生存数据

序号	x_1	x_2	x_3	x_4	x_5	y	序号	x_1	x_2	x_3	x_4	x_5	y
1	70	64	5	1	1	1	21	60	37	13	1	1	0
2	60	63	9	1	1	0	22	90	54	12	1	0	1
3	70	65	11	1	1	0	23	50	52	8	1	0	1
4	40	69	10	1	1	0	24	70	50	7	1	0	1
5	40	63	58	1	1	0	25	20	65	21	1	0	0
6	70	48	9	1	1	0	26	80	52	28	1	0	1
7	70	48	11	1	1	0	27	60	70	13	1	0	0
8	80	63	4	2	1	0	28	50	40	13	1	0	0
9	60	63	14	2	1	0	29	70	36	22	2	0	0
10	30	53	4	2	1	0	30	40	44	36	2	0	0
11	80	43	12	2	1	0	31	30	54	9	2	0	0
12	40	55	2	2	1	0	32	30	59	87	2	0	0
13	60	66	25	2	1	1	33	40	69	5	3	0	0



例 2.11 建立 Logistic 模型

$$K_0^2 = 16.5948,$$

全局回归
关系显著

$$\ln\left(\frac{p_k}{1-p_k}\right) = \beta_0 + \sum \beta_k X_k$$

$$p_0 = P_{H_0}(K^2 \geq K_0^2) = P(\chi^2(5) > 16.5948) = 0.0053,$$

表 2.13 参数的最大似然估计及 Wald 检验结果

参数	估计值	标准差	Wald 值	p 值
β_0	-7.011 4	4.475 3	2.454 4	0.117 2
β_1	0.099 9	0.043 0	5.391 0	0.020 2
β_2	0.014 2	0.047 0	0.090 8	0.763 1
β_3	0.017 5	0.054 6	0.102 7	0.748 6
β_4	-1.083 0	0.587 2	3.401 3	0.065 1
β_5	-0.613 1	0.960 7	0.407 3	0.523 3

$$P(Y=1)$$

$$= \frac{\exp(-7.0114 + 0.0999X_1 + 0.0142X_2 + 0.0175X_3 - 1.0830X_4 - 0.6131X_5)}{1 + \exp(-7.0114 + 0.0999X_1 + 0.0142X_2 + 0.0175X_3 - 1.0830X_4 - 0.6131X_5)},$$

基于向后剔除法选择模型的参数估计

表 2.14 基于向后剔除法选择模型的参数估计结果

参数	估计值	标准差	Wald 值	p 值
β_0	-6.137 5	2.738 4	5.023 2	0.025 0
β_1	0.097 6	0.040 8	5.724 1	0.016 7
β_4	-1.125 2	0.602 4	3.489 2	0.061 8

$$P(Y=1) = \frac{\exp(-6.137\ 5 + 0.097\ 6X_1 - 1.125\ 2X_4)}{1 + \exp(-6.137\ 5 + 0.097\ 6X_1 - 1.125\ 2X_4)}, \quad (2.100)$$



拟合值与观测值比较

表 2.15 40 位患者生存概率 $P(Y=1)$ 的拟合值及 Y 的观测值

序号	$\hat{P}(Y=1)$	y	序号	$\hat{P}(Y=1)$	y	序号	$\hat{P}(Y=1)$	y	序号	$\hat{P}(Y=1)$	y
1	0.393 7 *	1	11	0.358 7	0	21	0.196 6	0	31	0.004 2	0
2	0.196 6	0	12	0.011 2	0	22	0.820 5	1	32	0.004 2	0
3	0.393 7	0	13	0.073 6 *	1	23	0.084 4 *	1	33	0.003 7	0
4	0.033 6	0	14	0.011 2	0	24	0.393 7 *	1	34	0.025 1	0
5	0.033 6	0	15	0.000 5	0	25	0.004 9	0	35	0.153 6	0
6	0.393 7	0	16	0.009 6	0	26	0.632 8	1	36	0.666 8 *	0
7	0.393 7	0	17	0.221 3	0	27	0.196 6	0	37	0.038 8	0
8	0.358 7	0	18	0.096 7	0	28	0.084 4	0	38	0.429 9	0
9	0.073 6	0	19	0.841 5	1	29	0.174 1	0	39	0.841 5	1
10	0.004 2	0	20	0.666 8	1	30	0.011 2	0	40	0.666 8	1

只有1, 13, 23, 24, 36有误, 估计正确率80%

预测: 新患者数据, $x_1=85$, $x_4=1$

$$P(Y=1) = \frac{\exp(-6.137\ 5 + 0.097\ 6 \times 85 - 1.125\ 2 \times 1)}{1 + \exp(-6.137\ 5 + 0.097\ 6 \times 85 - 1.125\ 2 \times 1)} = 0.737\ 9.$$



卡方分布表

卡方分布表

df/a	0.995	0.990	0.975	0.950	0.900	0.750	0.250	0.100	0.050	0.025	0.010	0.005
1	3.9E-05	1.6E-04	9.8E-04	3.9E-03	0.016	0.102	1.323	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	0.575	2.773	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	1.213	4.108	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	1.923	5.385	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	2.675	6.626	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	3.455	7.879	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	4.255	9.037	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	5.071	10.219	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	5.899	11.389	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	6.737	12.549	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	7.584	13.701	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	8.438	14.845	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	9.299	15.984	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	10.165	17.117	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	11.037	18.245	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	11.912	19.369	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	12.792	20.489	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	13.675	21.605	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	14.562	22.718	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	15.452	23.828	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	16.344	24.935	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	17.240	26.039	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	18.137	27.141	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	19.037	28.241	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	19.939	29.339	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	20.843	30.435	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	21.749	31.528	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	22.657	32.620	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	23.567	33.711	39.087	42.557	45.722	49.588	52.338
30	13.787	14.951	16.791	18.493	20.599	24.478	34.802	40.289	43.772	46.999	50.892	53.672
31	14.450	15.651	17.539	19.281	21.432	25.390	35.893	41.479	44.984	48.278	52.196	55.005
32	15.110	16.355	18.291	20.072	22.267	26.303	36.984	42.656	46.153	49.588	53.490	56.328
33	15.768	17.063	19.047	20.866	23.103	27.217	38.075	43.828	47.331	50.892	54.784	57.641
34	16.424	17.775	19.806	21.663	23.940	28.132	39.166	44.984	48.502	52.196	56.078	58.954
35	17.078	18.490	20.568	22.462	24.778	29.047	40.257	46.153	49.660	53.490	57.372	60.259
36	17.730	19.208	21.332	23.263	25.617	29.962	41.348	47.331	50.805	54.784	58.666	61.563
37	18.380	19.929	22.098	24.066	26.457	30.877	42.439	48.502	51.942	56.078	59.960	62.867
38	19.028	20.652	22.866	24.871	27.298	31.792	43.530	49.660	53.074	57.372	61.254	64.171
39	19.674	21.377	23.635	25.678	28.139	32.707	44.621	50.805	54.181	58.666	62.558	65.475
40	20.318	22.103	24.405	26.486	28.980	33.622	45.712	51.942	55.289	59.960	63.862	66.779
41	20.960	22.830	25.176	27.295	29.821	34.537	46.803	53.074	56.396	61.254	65.166	68.083
42	21.601	23.558	25.948	28.105	30.662	35.452	47.894	54.181	57.507	62.558	66.470	69.387
43	22.241	24.286	26.720	28.915	31.503	36.367	48.985	55.289	58.618	63.862	67.774	70.691
44	22.880	25.015	27.493	29.725	32.344	37.282	50.076	56.396	59.729	65.166	69.078	71.995
45	23.518	25.744	28.265	30.535	33.185	38.197	51.167	57.507	60.839	66.470	70.382	73.299
46	24.156	26.473	29.037	31.345	34.026	39.112	52.258	58.618	61.942	67.774	71.686	74.603
47	24.793	27.202	29.809	32.155	34.867	40.027	53.349	59.729	63.053	69.078	72.990	75.907
48	25.429	27.931	30.581	32.965	35.708	40.942	54.440	60.839	64.164	70.382	74.294	77.211
49	26.065	28.660	31.353	33.775	36.549	41.857	55.531	61.942	65.275	71.686	75.598	78.515
50	26.700	29.389	32.125	34.585	37.390	42.772	56.622	63.053	66.386	72.990	76.902	79.819
51	27.335	30.118	32.897	35.395	38.231	43.687	57.713	64.164	67.497	74.294	78.206	81.123
52	27.969	30.847	33.669	36.205	39.072	44.602	58.804	65.275	68.608	75.598	79.510	82.427
53	28.603	31.576	34.441	37.015	39.913	45.517	59.895	66.386	69.719	76.902	80.814	83.731
54	29.237	32.304	35.213	37.825	40.754	46.432	60.986	67.497	70.830	78.206	82.118	85.035
55	29.870	33.033	35.985	38.635	41.595	47.347	62.077	68.608	71.941	79.510	83.422	86.339
56	30.503	33.762	36.757	39.445	42.436	48.262	63.168	69.719	73.052	80.814	84.726	87.643
57	31.136	34.491	37.529	40.255	43.277	49.177	64.259	70.830	74.163	82.118	86.030	88.947
58	31.769	35.220	38.301	41.065	44.118	50.092	65.350	71.941	75.274	83.422	87.334	90.251
59	32.401	35.948	39.073	41.875	44.959	51.007	66.441	73.052	76.385	84.726	88.638	91.555
60	33.034	36.677	39.845	42.685	45.800	51.922	67.532	74.163	77.496	86.030	89.942	92.859
61	33.666	37.405	40.617	43.495	46.641	52.837	68.623	75.274	78.607	87.334	91.246	94.163
62	34.298	38.134	41.389	44.305	47.482	53.752	69.714	76.385	79.718	88.638	92.550	95.467
63	34.930	38.862	42.161	45.115	48.323	54.667	70.805	77.496	80.829	89.942	93.854	96.771
64	35.562	39.591	42.933	45.925	49.164	55.582	71.896	78.607	81.940	91.246	95.158	98.075
65	36.194	40.319	43.705	46.735	50.005	56.497	72.987	79.718	83.051	92.550	96.462	99.379
66	36.826	41.048	44.477	47.545	50.846	57.412	74.078	80.829	84.162	93.854	97.766	100.683
67	37.458	41.776	45.249	48.355	51.687	58.327	75.169	81.940	85.273	95.158	99.070	101.987
68	38.090	42.505	46.021	49.165	52.528	59.242	76.260	83.051	86.384	96.462	100.374	103.291
69	38.722	43.233	46.793	49.975	53.369	60.157	77.351	84.162	87.495	97.766	101.678	104.595
70	39.354	43.962	47.565	50.785	54.210	61.072	78.442	85.273	88.606	99.070	102.982	105.899
71	39.986	44.690	48.337	51.595	55.051	61.987	79.533	86.384	89.717	100.374	104.286	107.203
72	40.618	45.419	49.109	52.405	55.892	62.902	80.624	87.495	90.828	101.678	105.590	108.507
73	41.250	46.147	49.881	53.215	56.733	63.817	81.715	88.606	91.939	102.982	106.894	109.811
74	41.882	46.876	50.653	54.025	57.574	64.732	82.806	89.717	93.050	104.286	108.198	111.115
75	42.514	47.604	51.425	54.835	58.415	65.647	83.897	90.828	94.161	105.590	109.502	112.419
76	43.146	48.333	52.197	55.645	59.256	66.562	84.988	91.939	95.272	106.894	110.806	113.723
77	43.778	49.061	52.969	56.455	60.097	67.477	86.079	93.050	96.383	108.198	112.110	115.027
78	44.410	49.790	53.741	57.265	60.938	68.392	87.170	94.161	97.494	109.502	113.414	116.331
79	45.042	50.518	54.513	58.075	61.779	69.307	88.261	95.272	98.605	110.806	114.718	117.635
80	45.674	51.247	55.285	58.885	62.620	70.222	89.352	96.384	99.716	112.110	116.022	118.939
81	46.306	51.975	56.057	59.695	63.461	71.137	90.443	97.495	100.827	113.414	117.326	120.243
82	46.938	52.704	56.829	60.505	64.302	72.052	91.534	98.606	101.938	114.718	118.630	121.547
83	47.570	53.432	57.601	61.315	65.143	72.967	92.625	99.717	103.049	116.022	119.934	122.851
84	48.202	54.161	58.373	62.125	65.984	73.882	93.716	100.828	104.160	117.326	121.238	124.155
85	48.834	54.889	59.145	62.935	66.825	74.797	94.807	101.939	105.271	118.630	122.542	125.459
86	49.466	55.618	59.917	63.745	67.666	75.7						

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□ 习题2.1

□ 习题2.2 - (1), (2)

